SCOTTISH QUALIFICATIONS AUTHORITY

2006 Mathematics

Higher − Paper 2

Finalised Marking Instructions

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1. Marks must be assigned in accordance with these marking instructions. In principle, marks are awarded for what is correct, rather than marks deducted for what is wrong.

2. Award one mark for each 'bullet' point. Each error should be underlined in RED at the point in the working where it first occurs, and not at any subsequent stage of the working.

3. The working subsequent to an error must be followed through by the marker with possible full marks for the subsequent working, provided that the difficulty involved is approximately similar. Where, subsequent to an error, the working is eased, a deduction(s) of mark(s) should be made.
   This may happen where a question is divided into parts. In fact, failure to even answer an earlier section does not preclude a candidate from assuming the result of that section and obtaining full marks for a later section.

4. Correct working should be ticked (✓). This is essential for later stages of the SQA procedures. Where working subsequent to an error(s) is correct and scores marks, it should be marked with a crossed tick (✗ or ✗✓). In appropriate cases attention may be directed to work which is not quite correct (e.g. bad form) but which has not been penalised, by underlining with a dotted or wavy line.
   Work which is correct but inadequate to score any marks should be corrected with a double cross tick (✗✗).

5. • The total mark for each section of a question should be entered in red in the **outer** right hand margin, opposite the end of the working concerned.
   • Only the mark should be written, **not** a fraction of the possible marks.
   • These marks should correspond to those on the question paper and these instructions.

6. It is of great importance that the utmost care should be exercised in adding up the marks. Where appropriate, all summations for totals and grand totals must be carefully checked. Where a candidate has scored zero marks for any question attempted, "0" should be shown against the answer.

7. As indicated on the front of the question paper, full credit should only be given where the solution contains appropriate working. Accept answers arrived at by inspection or mentally where it is possible for the answer so to have been obtained. Situations where you may accept such working will normally be indicated in the marking instructions.

8. Do not penalise:
   - working subsequent to a correct answer
   - legitimate variations in numerical answers
   - correct working in the "wrong" part of a question
   - omission of units
   - bad form

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9. No piece of work should be scored through without careful checking - even where a fundamental misunderstanding is apparent early in the answer. Reference should always be made to the marking scheme - answers which are widely off-beam are unlikely to include anything of relevance but in the vast majority of cases candidates still have the opportunity of gaining the odd mark or two provided it satisfies the criteria for the mark(s).

10. If in doubt between two marks, give an intermediate mark, but without fractions. When in doubt between consecutive numbers, give the higher mark.

11. In cases of difficulty covered neither in detail nor in principle in the Instructions, attention may be directed to the assessment of particular answers by making a referal to the P.A. Please see the general instructions for P.A. referrals.

12. No marks should be deducted at this stage for careless or badly arranged work. In cases where the writing or arrangement is very bad, a note may be made on the upper left-hand corner of the front cover of the script.

13 Transcription errors: In general, as a consequence of a transcription error, candidates lose the opportunity of gaining either the first ic mark or the first pr mark.

14 Casual errors: In general, as a consequence of a casual error, candidates lose the opportunity of gaining the appropriate ic mark or pr mark.

15 **Do not write any comments on the scripts**. A **revised** summary of acceptable notation is given on page 4.

16 Working that has been crossed out by the candidate cannot receive any credit. If you feel that a candidate has been disadvantaged by this action, make a P.A. Referral.

17 Throughout this paper, unless specifically mentioned, a correct answer with no working receives no credit.

Summary

Throughout the examination procedures many scripts are remarked. It is essential that markers follow common procedures:

1 **Tick** correct working.
2 Put a mark in the **outer right-hand margin to match the marks allocations on the question paper.**
3 Do **no**t write marks as fractions.
4 Put each mark **at the end** of the candidate's response to the question.
5 **Follow through** errors to see if candidates can score marks subsequent to the error.
6 Do **not** write any comments on the scripts.

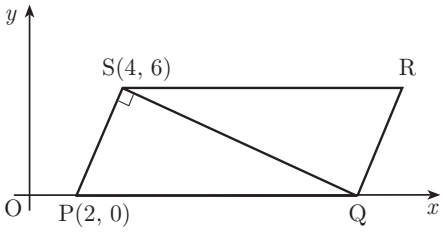
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1 PQRS is a parallelogram. P is the point (2, 0), S is (4, 6) and Q lies on the x-axis, as shown.

The diagonal QS is perpendicular to the side PS.

(a) Show that the equation of QS is $x + 3y = 22$. **4**

(b) Hence find the coordinates of Q and R. **2**



Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
1	a,b	4,2	C	G8	CN	06/05

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- •1 pr find gradient from two points
- •2 ss use $m_1 m_2 = -1$
- •3 ic state equation of the line
- •4 ic completes proof
- •5 ic interpret diagram
- •6 ic interpret diagram

Primary Method : Give 1 mark for each •

- •1 $m_{PS} = 3$
- •2 $m_{QS} = -\frac{1}{3}$
- •3 $y - 6 = -\frac{1}{3}(x - 4)$
- •4 completes proof **4 marks**
- •5 $Q = (22, 0)$
- •6 $R = (24, 6)$ **2 marks**

Notes

In (a)

1 In the Primary method, •3 is only available if an attempt has been made to find and use a perpendicular gradient.

2 In the Primary method and the Alt. method 1, •4 is only available for reaching the required equation.

3 To gain •4, some evidence of completion needs to be shown

e.g. $y - 6 = -\frac{1}{3}(x - 4)$

$3(y - 6) = -(x - 4)$

$x + 3y = 22$

4 Sometimes candidates manage to find R first. Provided the coordinates of R are of the form (?, 6), only then is •6 available as a follow through.

5 •5 and •6 are available to candidates who use their own erroneous equation for QS.

General Notes applicable throughout the marking scheme

There are many instances when follow throughs come into play and these will not always be highlighted for you. The following example is a reminder of what you have to look out for when you are marking.

example

At the •5 stage a candidate may switch the coordinates round so we have

•5 X $Q(0, 22)$

•6 $X\surd$ $R(2, 28)$ *repeated error*

so the candidate loses •5 for switching the coordinates but gains •6 as a consequence of following through.

Any error can be followed through and the subsequent marks awarded provided the working has not been eased.

Any deviation from this will be noted in the marking scheme.

Alternative Method 1

- •1 $m_{PS} = 3$
- •2 $m_{QS} = -\frac{1}{3}$

  $y = -\frac{1}{3}x + c$
- •3 $6 = -\frac{1}{3} \times 4 + c$
- •4 completes proof
- •5 $Q = (22, 0)$
- •6 $R = (24, 6)$

Alternative Method 2

Let $Q = (q, 0)$

- •1 $(q - 2)^2 = 2^2 + 6^2 + (q - 4)^2 + 6^2$
- •2 $q = 22$
- •3 $Q = (22, 0)$ *and* $R = (24, 6)$
- •4 $m_{QS} = -\frac{1}{3}$
- •5 $y - 0 = -\frac{1}{3}(x - 22)$
- •6 *leading to* $3y + x = 22$

N.B.

The coordinates of Q can also be arrived at by right-angled trig. Use the alt. method 2 marking scheme with •1 replaced by appropriate trig. work.

The only acceptable value for q is 22.

2 Find the value of k such that the equation $kx^2 + kx + 6 = 0$, $k \neq 0$, has equal roots. **4**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
2		4	C	A18	CN	06/new

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- $\bullet^1$ ss know to use "discriminant = 0"
- $\bullet^2$ ic interpret a,b,c
- $\bullet^3$ pr substitute & factorise
- $\bullet^4$ ic interpret solution

Primary Method : Give 1 mark for each •

- $\bullet^1$ $"b^2 - 4ac" = 0$
- $\bullet^2$ $a = k, b = k, c = 6$
- $\bullet^3$ $k(k - 24)$
- $\bullet^4$ $\begin{bmatrix} k = 0 \quad and \quad k = 24 \\ \therefore \quad k = 24 \end{bmatrix}$

4 marks

Notes

1 The evidence for $\bullet^1$ and/or $\bullet^2$ may not appear until the working immediately preceding the evidence for $\bullet^3$.
i.e. a candidate may simply start

$\surd \bullet^1, \surd \bullet^2 \quad k^2 - 4 \times k \times 6 = 0$
$\surd \bullet^3 \quad k(k - 24)$

or

$\surd \bullet^2 \quad k^2 - 4 \times k \times 6$
$\surd \bullet^1, \surd \bullet^3 \quad k(k - 24) = 0$

2 The "= 0" has to appear at least once, at the $\bullet^1$ stage or at the $\bullet^3$ stage.

3 In the Primary method, candidates who do not deal with the root k = 0 cannot obtain $\bullet^4$.
[see Common Errors 1 and 2]
Minimum evidence for $\bullet^4$ would be scoring out "k = 0" or "k = 24" underlined.

4 Some candidates may start with the quadratic formula. Apply the marking scheme to the part underneath the square root sign.

5 The use of any expression masquerading as the discriminant can only gain $\bullet^2$ at most.

Alternative Method 1 (completing the square)

- $\bullet^1$ $\left(x + \frac{1}{2}\right)^2 + \ldots\ldots$
- $\bullet^2$ $\left(x + \frac{1}{2}\right)^2 - \frac{1}{4} + \frac{6}{k} = 0$
- $\bullet^3$ $equal\ roots \Rightarrow -\frac{1}{4} + \frac{6}{k} = 0$
- $\bullet^4$ $k = 24$

Acceptable alternative for •4

$\surd$	$\bullet^1$	$"b^2 - 4ac" = 0$
$\surd$	$\bullet^2$	$a = k, b = k, c = 6$
$\surd$	$\bullet^3$	$k(k - 24)$
$\surd$	$\bullet^4$	$\cancel{k = 0}\ or\ 24$

Common Error 1 at the •4 stage

$\surd$	$\bullet^1$	$"b^2 - 4ac" = 0$
$\surd$	$\bullet^2$	$a = k, b = k, c = 6$
$\surd$	$\bullet^3$	$k(k - 24)$
X	$\bullet^4$	$k = 0\ or\ 24$

Common Error 2 at the •4 stage

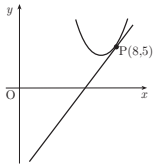
$\surd$	$\bullet^1$	$"b^2 - 4ac" = 0$
$\surd$	$\bullet^2$	$a = k, b = k, c = 6$
$\surd$	$\bullet^3$	$k(k - 24)$
X	$\bullet^4$	$k = 24$

Common Error 3 Division by k

$\surd$	$\bullet^1$	$"b^2 - 4ac" = 0$
$\surd$	$\bullet^2$	$a = k, b = k, c = 6$
X	$\bullet^3$	$k^2 - 24k = 0$
		$k^2 = 24k$
X	$\bullet^4$	$k = 24$

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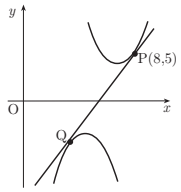
3 The parabola with equation $y = x^2 - 14x + 53$ has a tangent at the point P(8,5).



4

(a) Find the equation of this tangent.

(b) Show that the tangent found in (a) is also a tangent to the parabola with equation $y = -x^2 + 10x - 27$ and find the coordinates of the point of contact Q.



5

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
3	a	4	C	C5	CN	06/26
	b	5	C	A24	CN	

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- $\bullet^1$ ss know to differentiate
- $\bullet^2$ pr differentiate
- $\bullet^3$ pr evaluate gradient
- $\bullet^4$ ic state equation of tangent
- $\bullet^5$ ss arrange in standard form
- $\bullet^6$ ss substitute into quadratic
- $\bullet^7$ pr process
- $\bullet^8$ ic factorise & interpret
- $\bullet^9$ ic state coordinates

Primary Method : Give 1 mark for each •

- $\bullet^1$ $\dfrac{dy}{dx} =$
- $\bullet^2$ $2x - 14$
- $\bullet^3$ $m = 2$ **stated or implied by •4**
- $\bullet^4$ $y - 5 = 2(x - 8)$ **4 marks**

- $\bullet^5$ $y = 2x - 11$
- $\bullet^6$ $2x - 11 = -x^2 + 10x - 27$
- $\bullet^7$ $x^2 - 8x + 16 = 0$
- $\bullet^8$ $(x-4)^2 = 0 \Rightarrow$ *equal roots so tgt*
- $\bullet^9$ $\mathrm{Q} = (4,-3)$ **5 marks**

Notes

In (a)

1 $\bullet^4$ is only available if an attempt has been made to find the gradient from differentiation.

In (b)

2 $\bullet^6$ is only available for a numerical value of m.

3 An "= 0" must occur somewhere in the working between $\bullet^7$ and $\bullet^8$.

4 $\bullet^8$ is awarded for drawing a conclusion from the candidate's quadratic equation.

5 Candidates may substitute the equation of the parabola into the equation of the line. This is a perfectly acceptable approach.

Common Error 1

$\surd$	$\bullet^1$	$\dfrac{dy}{dx} =$
$\surd$	$\bullet^2$	$2x - 14$
X	$\bullet^3$	$2x - 14 = 0$ *so* $x = 7$ *so* $m = 7$
X	$\bullet^4$	$y - 5 = 7(x - 8)$
$X\surd$	$\bullet^5$	$y = 7x - 51$
$X\surd$	$\bullet^6$	$7x - 51 = -x^2 + 10x - 27$
$X\surd$	$\bullet^7$	$x^2 - 3x - 24 = 0$
$X\surd$	$\bullet^8$	$b^2 - 4ac = 105 \Rightarrow$ *line is not tgt*
X	$\bullet^9$	$--$

so award 6 marks

Alternative Marking 1 [Marks 8]

- $\bullet^8$ $b^2 - 4ac = 64 - 4 \times 16 = 0 \Rightarrow$ *line is a tangent*

Alternative Method 1 for (b)

- $\bullet^5$ $2x = y + 11$
- $\bullet^6$ $4y = -(y^2 + 22y + 121) + 20y + 220 - 108$
- $\bullet^7$ $y^2 + 6x + 9 = 0$
- $\bullet^8$ $(y+3)^2 = 0 \Rightarrow$ *equal roots so tgt*
- $\bullet^9$ $\mathrm{Q} = (4,-3)$

Alternative Method 2 for (b)

- $\bullet^5$ *Find the equ. of the tgt to 2nd curve with grad. 2*
  stated or implied by •6
- $\bullet^6$ $-2x + 10 = 2$
- $\bullet^7$ $\mathrm{Q} = (4,-3)$
- $\bullet^8$ $y - (-3) = 2(x - 4)$
- $\bullet^9$ $y = 2x - 11$ *which is the same equ. as* (a)
  stated explicitly

8

4 The circles with equations $(x-3)^2+(y-4)^2=25$ and $x^2+y^2-kx-8y-2k=0$.
have the same centre. Determine the radius of the larger circle. **5**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
4		5	C	G9	CN	06/55

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- $\bullet^1$ ic state centre of circle 1
- $\bullet^2$ ss equate x-coordinates, find k.
- $\bullet^3$ ic find radius of circle 1
- $\bullet^4$ ic substitute into the radius formula
- $\bullet^5$ ic process radius formula and compare.

Primary Method : Give 1 mark for each •

- $\bullet^1$ $C_1=(3,4)$
- $\bullet^2$ $k=6$
- $\bullet^3$ $R_1=5$
- $\bullet^4$ $R_2=\sqrt{(-3)^2+(-4)^2-(-12)}$ **or equivalent**
- $\bullet^5$ $\sqrt{37}>5$ *or* "2nd circle" **5 marks**

Notes

1 $\bullet^2$ requires no justification.

2 Evidence for $\bullet^3$ may appear for the first time at the $\bullet^5$ stage.

3 If $R_1 = 5$ is clearly stated at the $\bullet^3$ stage, then it does not have to appear at the $\bullet^5$ stage for the conclusion to be drawn.

4 For any formula masquerading as the radius formula (e.g. see Common Error 2) , $\bullet^4$ and $\bullet^5$ are NOT available.

Alternative Method 1

- $\bullet^1$ $x^2+y^2-6x-8y+25=25$
- $\bullet^2$ $k=6$
- $\bullet^3$ $R_1=5$
- $\bullet^4$ $R_2=\sqrt{(-3)^2+(-4)^2-(-12)}$ **or equivalent**
- $\bullet^5$ $\sqrt{37}>5$ *or* "2nd circle"

Common Error 1

$\surd$	$\bullet^1$	$C_1=(3,4)$
$\surd$	$\bullet^2$	$k=6$
$\surd$	$\bullet^3$	$R_1=5$
X	$\bullet^4$	$R_2=\sqrt{(-3)^2+(-4)^2-12}$
$X\surd$	$\bullet^5$	$\sqrt{13}<5$ *or* "1st circle"

Common Error 2

$\surd$	$\bullet^1$	$C_1=(3,4)$
$\surd$	$\bullet^2$	$k=6$
$\surd$	$\bullet^3$	$R_1=5$
X	$\bullet^4$	$R_2=\sqrt{(-3)^2+(-4)^2+(12)^2}$
X	$\bullet^5$	$13>5$ *or* "2nd circle"

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5 The curve $y = f(x)$ is such that $\frac{dy}{dx} = 4x - 6x^2$. The curve passes through the point $(-1, 9)$. Express y in terms of x. **4**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
5		4	C/B	C18	CN	06/37

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- •1 ss know to integrate
- •2 pr integrate
- •3 ic substitute values
- •4 pr process constant

Primary Method : Give 1 mark for each •

- •1 $y = \int ...$ **stated or implied by •2**
- •2 $\frac{4}{2}x^2 - \frac{6}{3}x^3$
- •3 $9 = 2(-1)^2 - 2(-1)^3 + c$
- •4 $y = 2x^2 - 2x^3 + 5$ **stated explicitly** **4 marks**

Notes

1 The equation "y = ..." must appear somewhere in the solution.

Common Error 1 Missing "equation"

√	•1	$y = \int ...$
√	•2	$\frac{4}{2}x^2 - \frac{6}{3}x^3$
√	•3	$9 = 2(-1)^2 - 2(-1)^3 + c$
X	•4	$c = 5$

award 3 marks

Common Error 2 : Not using (–1, 9)

√	•1	$y = \int ..$
√	•2	$\frac{4}{2}x^2 - \frac{6}{3}x^3$
X	•3	$2(-1)^2 - 2(-1)^3 + c = 0$
X	•4	$y = 2x^2 - 2x^3 - 4$

award 2 marks

Alternative Marking

- •1 $y = \int ...$
- •2 $\frac{4}{2}x^2 - \frac{6}{3}x^3$
- •3 $y = 2x^2 - 2x^3 + c$ *and* $9 = 2(-1)^2 - 2(-1)^3 + c$ **stated explicitly**
- •4 $c = 5$

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6 P is the point (–1, 2, –1) and Q is (3, 2, –4).

(*a*) Write down $\overrightarrow{PQ}$ in component form. **1**

(*b*) Calculate the length of $\overrightarrow{PQ}$. **1**

(*c*) Find the components of a unit vector which is parallel to $\overrightarrow{PQ}$. **1**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
6	a	1	C	G17	CN	06/59
	b	1	C	G16		
	c	1	B	G18		

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- •1 ic state vector components
- •2 pr find the length of a vector
- •3 ic state unit vector

Primary Method : Give 1 mark for each •

- •1 $\overrightarrow{PQ} = \begin{pmatrix} 4 \\ 0 \\ -3 \end{pmatrix}$ **1 mark**
- •2 $|\overrightarrow{PQ}| = 5$ **1 mark**
- •3 $\begin{pmatrix} \frac{4}{5} \\ 0 \\ -\frac{3}{5} \end{pmatrix}$ **1 mark**

Note

In (a)

1 It is perfectly acceptable to write the components as a row vector eg $\overrightarrow{PQ} = \begin{pmatrix} 4 & 0 & -3 \end{pmatrix}$.

Treat $\overrightarrow{PQ} = (4, 0, -3)$ as bad form (i.e. not penalised).

In (b)

2 •2 is not awarded for an unsimplified $\sqrt{25}$.

3 Beware of misappropriate use of the scalar product where, by coincidence, $\boldsymbol{p}.\boldsymbol{q} = 5$.

In (c)

4 Accept $\frac{1}{5}\begin{pmatrix} 4 \\ 0 \\ -3 \end{pmatrix}$ for •3.

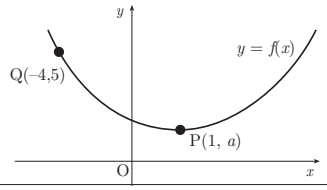
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7 The diagram shows the graph of a function $y = f(x)$.

Copy the diagram and on it sketch the graphs of

(a) $y = f(x-4)$ **2**

(b) $y = 2 + f(x-4)$ **2**



Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
7	a	2	C	A3	CN	06/new
	b	2	C	A3		

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- $\bullet^1$ ic know translate parallel to x-axis, +ve dir.
- $\bullet^2$ ic annotate points
- $\bullet^3$ ic know translate parallel to y-axis, +ve dir.
- $\bullet^4$ ic annotate points

Primary Method : Give 1 mark for each •

- $\bullet^1$ translate 4 units right and annotate one point
- $\bullet^2$ annotate the other point $[\mathrm{P}'(5,a)\ \mathrm{Q}'(0,5)]$

2 marks

- $\bullet^3$ translate (a) 2 units up and annotate one point
- $\bullet^4$ annotate the other point $[\mathrm{P}''(5,a+2)\ \mathrm{Q}''(0,7)]$

2 marks

Notes

For (a)

1 A translation of $\begin{pmatrix}-4\\0\end{pmatrix}$ earns a maximum of 1 mark with both points clearly annotated and f(x) retaining its shape.

2 Any other translation gains no marks.

In the Primary method

For (b)

3 $\bullet^3$ and $\bullet^4$ are only available for applying the translation to the resultant graph from (a).

4 A translation of $\begin{pmatrix}0\\-2\end{pmatrix}$ earns a maximum of 1 mark with both points clearly annotated and the resultant graph from (a) retaining its shape.

5 Any other translation gains no marks.

In the Alternative method

For (b)

6 A translation of $\begin{pmatrix}4\\-2\end{pmatrix}, \begin{pmatrix}-4\\2\end{pmatrix}$ *or* $\begin{pmatrix}-4\\-2\end{pmatrix}$ applied to the original graph earns a maximum of 1 mark with both points clearly annotated and the resultant graph retaining its original shape.

7 Any other translation gains no marks.

In either method

For (a) and (b)

8 For the annotated points, accept a superimposed grid or clearly labelled axes.

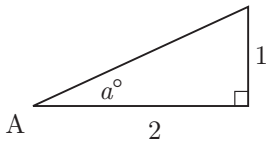
9 A candidate may choose to use two separate diagrams. This is acceptable.

Alternative Method

- $\bullet^1$ translate 4 units right and annotate one point
- $\bullet^2$ annotate the other point $[\mathrm{P}'(5,a)\ \mathrm{Q}'(0,5)]$
- $\bullet^3$ translate original $\begin{pmatrix}4\\2\end{pmatrix}$ and annotate one point
- $\bullet^4$ annotate the other point $[\mathrm{P}''(5,a+2)\ \mathrm{Q}''(0,7)]$

12

8 The diagram shows a right-angled triangle with height 1 unit, base 2 units and an angle of $a°$ at A.



(a) Find the exact values of
  (i) $\sin a°$
  (ii) $\sin 2a°$. **4**

(b) By expressing $\sin 3a°$ as $\sin(2a + a)°$, find the exact value of $\sin 3a°$. **4**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
8	a	4	C	T9	CN	06/44
	b	4	B	T8	CN	

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- •1 ic interpret diagram for $\sin(a°)$
- •2 ss use double angle formula for $\sin(2A)$
- •3 ic interpret diagram for $\cos(a°)$
- •4 pr substitute and complete
- •5 ss use compound angle formula
- •6 pr use double angle formula for $\cos(2A)$
- •7 ic substitute
- •8 pr complete

Primary Method : Give 1 mark for each •

•1 $\sin(a°) = \frac{1}{\sqrt{5}}$

•2 $\sin(2a°) = 2\sin(a°)\cos(a°)$

•3 $\cos(a°) = \frac{2}{\sqrt{5}}$

•4 $\sin(2a°) = \frac{4}{5}$ **4 marks**

•5 $\sin(3a°) = \sin(2a°)\cos(a°) + \cos(2a°)\sin(a°)$

•6 $\cos(2a°) = \frac{3}{5}$

•7 $\sin(3a°) = \frac{4}{5}.\frac{2}{\sqrt{5}} + \frac{3}{5}.\frac{1}{\sqrt{5}}$

•8 $\sin(3a°) = \frac{11}{5\sqrt{5}}$ **4 marks**

Note

1 Calculating approximate angles using arcsin and arccos gains no credit.

2 There are 3 processing marks •4, •6 and •8. None of these are available for an answer > 1.

3 sin(2a) = 0.8 and cos(2a) = 0.6 are the only two decimal fractions which may receive any credit.

4 Some candidates may double the height of the triangle and then call the base angle 2a. This error is equivalent to Common Error 1 illustrated on the right.

Common Error 2
An example based on a numerical error in Pythagoras

X	•1	$\sin(a°) = \frac{1}{\sqrt{3}}$
√	•2	$\sin(2a°) = 2\sin(a°)\cos(a°)$
X √	•3	$\cos(a°) = \frac{2}{\sqrt{3}}$
X	•4	$\sin(2a°) = \frac{4}{3}$
√	•5	$\sin(3a°) = \sin(2a°)\cos(a°) + \cos(2a°)\sin(a°)$
X	•6	$\cos(2a°) = 2\cos^2(a°) - 1 = \frac{5}{3}$ *or equivalent*
X √	•7	$\sin(3a°) = \frac{4}{3}.\frac{2}{\sqrt{3}} + \frac{5}{3}.\frac{1}{\sqrt{3}}$
X	•8	$\sin(3a°) = \frac{13}{3\sqrt{3}}$

Common Error 1 An example of Incorrect formulae

√	•1	$\sin(a°) = \frac{1}{\sqrt{5}}$
X	•2	$\sin(2a°) = 2\sin(a°)$
X	•4	$\sin(2a°) = \frac{2}{\sqrt{5}}$
√	•5	$\sin(3a°) = \sin(2a°)\cos(a°) + \cos(2a°)\sin(a°)$
√	•3	$\cos(a°) = \frac{2}{\sqrt{5}}$
X	•6	$\cos(2a°) = \frac{4}{\sqrt{5}}$
X √	•7	$\sin(3a°) = \frac{2}{\sqrt{5}}.\frac{2}{\sqrt{5}} + \frac{4}{\sqrt{5}}.\frac{1}{\sqrt{5}}$
X	•8	$\sin(3a°) = \frac{8}{5}$

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9 $y = \dfrac{1}{x^3} - \cos 2x,\ x \neq 0,$ find $\dfrac{dy}{dx}$. **4**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
8		4	C/B	C3,C20	CN	06/79

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- $\bullet^1$ ss express in differentiable form
- $\bullet^2$ pr differentiate a term with a negative power
- $\bullet^3$ pr start to process a compound derivative
- $\bullet^3$ pr complete compound derivative

Primary Method : Give 1 mark for each •

- $\bullet^1$ x^{-3}
- $\bullet^2$ $-3x^{-4}$
- $\bullet^3$ $+\sin 2x$
- $\bullet^4$ $\times 2$

4 marks

Notes

1 For clearly integrating, correctly or otherwise, only $\bullet^1$ is available.

2 If you cannot decide whether a candidate has attempted to differentiate or integrate, assume they have attempted to differentiate.

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10 A curve has equation $y = 7\sin x - 24\cos x$.

(a) Express $7\sin x - 24\cos x$ in the form $k\sin(x-a)$ where $k > 0$ and $0 \le a \le \frac{\pi}{2}$. **4**

(b) Hence find, in the interval $0 \le x \le \pi$, the x-coordinate of the point on the curve where the gradient is 1. **3**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
10	a	4	C	T13	CR	06/97
	b	3	A/B	T17	CR	

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- $\bullet^1$ ss expand
- $\bullet^2$ ic compare coefficients
- $\bullet^3$ pr process k
- $\bullet^4$ pr process a
- $\bullet^5$ ic state result
- $\bullet^6$ ss set derivative = gradient
- $\bullet^7$ pr process 'x' from the derivative

Primary Method : Give 1 mark for each •

$\bullet^1$	$k\sin(x)\cos(a) - k\cos(x)\sin(a)$	**stated explicitly**
$\bullet^2$	$k\cos(a) = 7,\ k\sin(a) = 24$	**stated explicitly**
$\bullet^3$	$k = 25$	
$\bullet^4$	$a = 1.29$	**4 marks**
$\bullet^5$	$25\sin(x - 1.29)$	
$\bullet^6$	$\frac{dy}{dx} = 25\cos(x - 1.29) = 1$	
$\bullet^7$	$x = 2.82$	**3 marks**

Notes

In (a)

1 $k\Big(\sin(x)\cos(a) - \cos(x)\sin(a)\Big)$ is acceptable for $\bullet^1$.

2 Treat $k\sin(x)\cos(a) - \cos(x)\sin(a)$ as bad form if $\bullet^2$ is gained.

3 No justification is required for $\bullet^3$.

4 $\bullet^3$ is not available for an unsimplified $\sqrt{625}$.

5 $25\Big(\sin(x)\cos(a) - \cos(x)\sin(a)\Big)$ is acceptable evidence for $\bullet^1$ and $\bullet^3$.

6 Candidates may use any form of the wave equation to start with as long as their final answer is in the form $k\sin(x-a)$. If it is not, then $\bullet^4$ is not available.

7 $\bullet^4$ is only available for
(i) an answer in radians which rounds to 1.3 OR
(ii) an answer given as a multiple of π *e.g.* $\frac{37}{90}\pi$.

8 $k\cos(a) = 7$ *and* $k\sin(a) = -24$ leading to a = 4.99 can only gain $\bullet^4$ if a comment intimating that this answer is not in the given interval is given.

In (b)

9 In (b) candidates have a choice of two starting points.
They can either start from $y = 25\sin(x - 1.29)$ as shown in the Primary method OR
they can start from $\frac{dy}{dx} = 7\cos(x) + 24\sin(x)$. Either of these starting positions may be awarded $\bullet^5$.

Common Error 1 Working in degrees

$\surd$	$\bullet^1$	$25\Big(\sin(x)\cos(a) - \cos(x)\sin(a)\Big)$
$\surd$	$\bullet^2$	$k\cos(a) = 7,\ k\sin(a) = 24$
$\surd$	$\bullet^3$	$k = 25$
X	$\bullet^4$	$a = 73.7$
$\surd$	$\bullet^5$	$25\sin(x - 73.7)$
X	$\bullet^6$	$\frac{dy}{dx} = 25\cos(x - 73.7) = 1$
$\surd$X	$\bullet^7$	$x = 161.4$

Award (a) 3 marks and (b) 2 marks

10 Candidates who work in degrees will lose $\bullet^6$ for attempting to differentiate .

11 $\bullet^7$ is only available as a consequence of solving $\frac{dy}{dx} = 1$. Do not penalise "extra" solutions at the $\bullet^7$ stage (e.g. 6.04). **15**

11 It is claimed that a wheel is made from wood which is over 1000 years old.
To test this claim, carbon dating is used.
The formula $A(t) = A_0 e^{-0.000124t}$ is used to determine the age of the wood, where A_0 is the amount of carbon in any living tree, $A(t)$ is the amount of carbon in the wood being dated and t is the age of the wood in years For the wheel it was found that $A(t)$ was 88% of the amount of carbon in a living tree.
Is the claim true?
5

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
11		5	A/B	A30	CR	06/36

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- $\bullet^1$ ic interpret information
- $\bullet^2$ ic substitute
- $\bullet^3$ ss take logarithms
- $\bullet^4$ pr process
- $\bullet^5$ ic interpret result

Primary Method : Give 1 mark for each •

- $\bullet^1$ $A(t) = 0.88A_0$ **stated or implied by $\bullet^2$**
- $\bullet^2$ $e^{-0.000124t} = 0.88$
- $\bullet^3$ $\ln\left(e^{-0.000124t}\right) = \ln\left(0.88\right)$ **stated or implied by $\bullet^4$**
- $\bullet^4$ $-0.000124t = \ln(0.88)$
- $\bullet^5$ $t = 1031$ years so claim valid **5 marks**

Notes

1 Candidates may choose a numerical value for A_0 at the start of their solution. Accept this situation.

2 $\bullet^5$ is only available if $\bullet^4$ has been awarded.

3 In following through from an error, $\bullet^5$ is only available for a positive value of t.

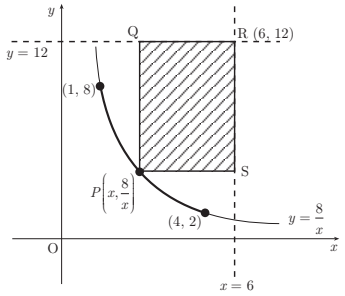
Alternative Method 1 Graph and Calculator Solution

- $\bullet^1$ $A(1000) = A_0 e^{-0.000124\times1000}$
- $\bullet^2$ $0.883A_0$ *and* 1000 year old piece of wood contains 88.3% carbon.
- $\bullet^3$ try a point where $t > 1030$
  e.g. $A(1050)$ getting $0.878A_0$
- $\bullet^4$ sketch of $y=A_0e^{-0.000124t}$ showing
  1. a monotonic decreasing function
  2. points representing eg (1000, 88.3%) etc
- $\bullet^5$ observation that the point lies between the two plotted values for t and so claim valid.

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12 PQRS is a rectangle formed according to the following conditions:

- it is bounded by the lines $x = 6$ and $y = 12$
- P lies on the curve with equation $y = \frac{8}{x}$ between (1, 8) and (4, 2)
- R is the point (6, 12).



(a) (i) Express the lengths of PS and RS in terms of x, the x-coordinate of P.

(ii) Hence show that the area, A square units, of PQRS is given by $A = 80 - 12x - \frac{48}{x}$. **3**

(b) Find the greatest and least possible values of A and the corresponding values of x for which they occur. **8**

Qu.	part	marks	Grade	Syllabus Code	Calculator class	Source
12	a	3	A	C12	CN	06/20
	b	9	A/B	C12		

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- $\bullet^1$ ic interpret diagram to find PS
- $\bullet^2$ ic interpret diagram to find RS
- $\bullet^3$ ic complete proof
- $\bullet^4$ ic express in differentiable form
- $\bullet^5$ ss know to set derivative to zero
- $\bullet^6$ pr differentiate
- $\bullet^7$ pr process equation
- $\bullet^8$ pr evaluate area at the turning point
- $\bullet^9$ pr evaluate area at the end point
- $\bullet^{10}$ pr evaluate area at the end point
- $\bullet^{11}$ ic state conclusion

Primary Method : Give 1 mark for each •

- $\bullet^1$ $PS = 6 - x$
- $\bullet^2$ $RS = 12 - \frac{8}{x}$
- $\bullet^3$ $Area = \left(6 - x\right)\left(12 - \frac{8}{x}\right)$ and complete **3 marks**
- $\bullet^4$ $48x^{-1}$
- $\bullet^5$ $\frac{dA}{dx} = 0$
- $\bullet^6$ $-12 + 48x^{-2}$
- $\bullet^7$ $x = 2$
- $\bullet^8$ $A(2) = 32$
- $\bullet^9$ $A(1) = 20$
- $\bullet^{10}$ $A(4) = 20$
- $\bullet^{11}$ $\max.A = 32 \ at \ x = 2$ ***and*** $\min.A = 20 \ at \ x = 1 \ or \ x = 4$ **8 marks**

Notes

1 For $\bullet^3$ there needs to be clear evidence that candidates have multiplied out the brackets in order to complete the proof.

2 An " = 0 " must appear somewhere in the working between $\bullet^4$ and $\bullet^7$.

3 At the $\bullet^7$ stage, ignore the omission or inclusion of x = –2.

4 $\bullet^8$ has to be as a consequence of solving $\frac{dA}{dx} = 0$.

5 $\bullet^{11}$ is only available if both end points have been considered.